

CNS Infections: Bacteria

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Make.
Heal.

Innovate.

Reinvent the Future.

Learning Objectives

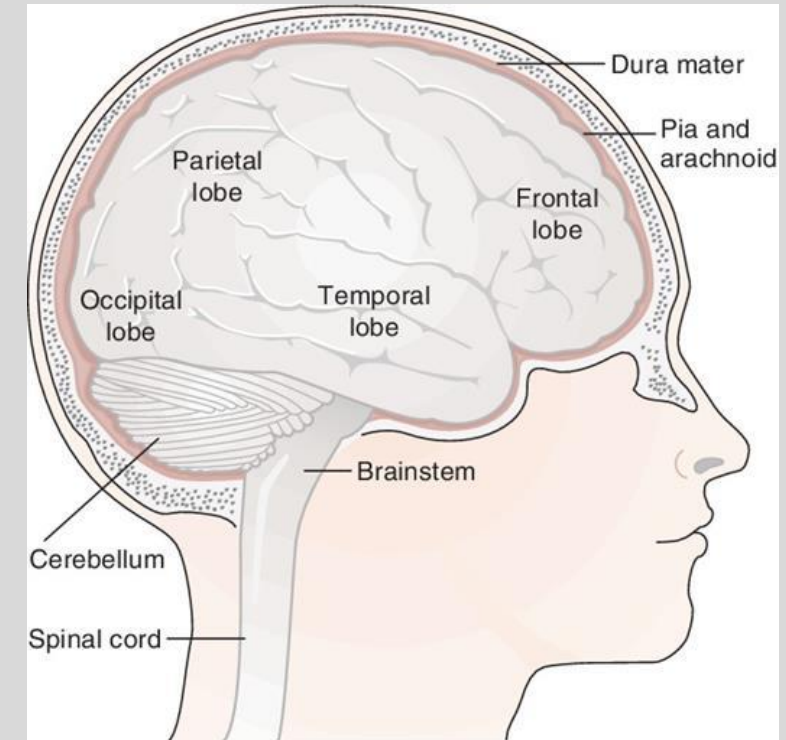
1. Describe the different types of bacterial CNS infections and their etiology
2. Compare and contrast bacterial and viral meningitis
3. Compare the properties of the most common bacterial etiological agents of meningitis
4. Identify the importance of diagnostic CSF analysis and antimicrobial therapy
5. Illustrate the role the patient age plays in determination of bacterial etiology

Outline

- Inflammation of the CNS (meningitis, encephalitis, myelitis)
- How CNS infection occurs
- How tissue is damaged in bacterial CNS infections
- Bacterial vs. Viral meningitis
- Acute bacterial meningitis, its epidemiology, and its etiologies
- Chronic bacterial meningitis and its etiologies
- Brain abscesses and their etiologies

Inflammation of the CNS

- Meningitis (meninges)
 - Acute or chronic
 - Bacterial, viral, fungal, or aseptic
 - Encephalitis (parenchyma)
 - Brain abscesses
 - Myelitis (spinal cord)
-
- Why can inflammation of the CNS result in such serious conditions?



How CNS Infection Occurs

- Entry to the CNS
 - Hematogenous from inoculation at another site
 - Respiratory epithelium (aerosol, droplets; e.g. meningococci)
 - Animal or bug bites
 - Transplacentally
 - **from blood to CNS: choroid plexus or infection of CNS endothelial cells
 - Direct inoculation
 - Trauma
 - Surgery
- + evasion of the immune system...

How CNS Infection Occurs

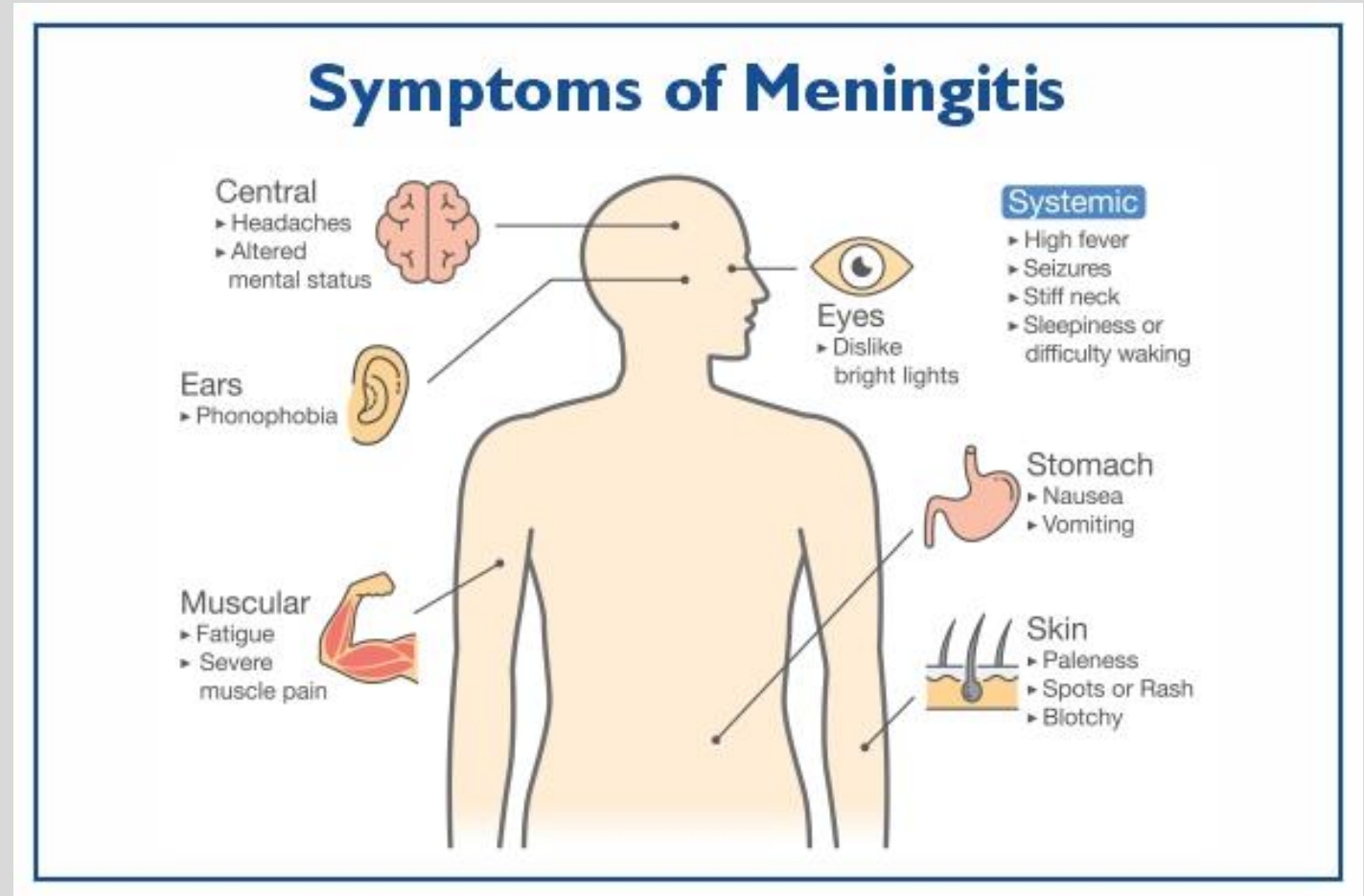
- Evading the immune system (i.e., virulence factors)
 - **Capsules** – sialic acid, adherence to meninges
 - K1 polysaccharide in *E. coli*
 - IgA protease, hemolysis, evading complement which is low in CSF
 - Children under 1 year
 - Immunocompromised
 - High microbe load
- Immune response of the CNS
 - Blood brain barrier
 - Microglia
 - Virchow-Robin spaces

How Tissue is Damaged in Bacterial CNS Infections

- Cell death from infection (toxins, lysis)
- Inflammatory response
 - Cytokines, chemokines
 - Capillary permeability
 - Phagocytes release enzymes
 - Edema
 - Brain and brainstem herniation
 - Decreased perfusion of fluids (venous backup)

How Tissue is Damaged in Bacterial CNS Infections

- Symptoms occur from:
 - Direct tissue damage
 - Increased pressure
 - Displacing of structures
- Focal or Global



Bacterial vs. Viral Meningitis

Acute Meningitis

	Bacterial	Viral
Presentation	Focal or global	Mostly focal
CSF	↑ Opening pressure ↑ PMNs ↑ Protein ↓ Glucose	Normal/↑ Opening pressure ↑ Lymphocytes Normal protein Normal/↑ glucose
Outcomes	High mortality and morbidity	Low mortality and morbidity
Tx	Immediate abx therapy	Supportive therapy

Bacterial vs. Viral Meningitis

CSF Composition with numerical measurements

TABLE 61-4

Usual Composition of Cerebrospinal Fluid in Various Central Nervous System Infections

Factor	Normal	Acute Bacterial Meningitis	Chronic Mycobacterial and Fungal Meningitis	Acute Viral Meningitis	Viral Encephalitis	Brain Abscess
Leukocytes (per mm ³)	0–6	>1,000	100–500	10–1,000, usually <300	10–500	10–100
% Neutrophils	0	>50	<10	May predominate in first 24 h, then <50	May predominate in first 24 h, then <50	<50
Red blood cells (per mm ³)	0–2	0–10	0–2	0–2	10–500 (HSV)	0–2
Glucose (mg/dL) ^a	40–80	<30	≤40	40–80	40–80	40–80
Protein (mg/dL)	20–50	>100	50–100 (fungal), >100 (mycobacterial)	50–100	50–100	50–100

^aDiagnostic values are best interpreted as the ratio of glucose levels in CSF compared with those in serum. Ratios <0.4 are considered diminished.

Review Question 1

Choose from the options in parentheses to complete the sentences or fill in the blanks:

In a suspected case of meningitis, CSF containing high protein and low glucose may indicate (**bacterial/viral**) meningitis which requires _____ treatment.

In a suspected case of meningitis, CSF containing high lymphocytes and normal glucose may indicate (**bacterial/viral**) meningitis which requires _____ treatment.

Acute Bacterial Meningitis

- Acute – onset and progression is rapid, occurs in days
- Isolated infection or from septicemia
- Fever, stiff neck, focal or global CNS dysfunction, petechiae
 - Purulent infections (pus) leads to increased pressure > nausea, vomiting, CNS depression
- Etiology is age dependent
- Fatal without treatment

Acute Bacterial Meningitis

Etiologies of Acute Bacterial Meningitis – from First Aid and Schaechter’s

Newborn (0-6mo)	Children (6mo-60yr)	Elderly (60+yr)	Any age – cranial/dental surgery	Any age - immunocompromised
Group B Strep (<i>S. agalactiae</i>) <i>E. coli</i> <i>L. monocytogenes</i>	<i>S. pneumoniae</i> <i>N. meningitidis</i> <i>H. influenzae</i> type B	<i>S. pneumoniae</i> <i>L. monocytogenes</i> <i>N. meningitidis</i> Group B Strep (<i>S. agalactiae</i>) <i>H. influenzae</i> type B	<i>S. aureus</i> * <i>S. epidermidis</i> Gram negs (<i>P. aeruginosa</i>) * <i>S. aureus</i> in abscesses	<i>L. monocytogenes</i> Gram negs (<i>P. aeruginosa</i>) **TB in <u>chronic</u> meningitis

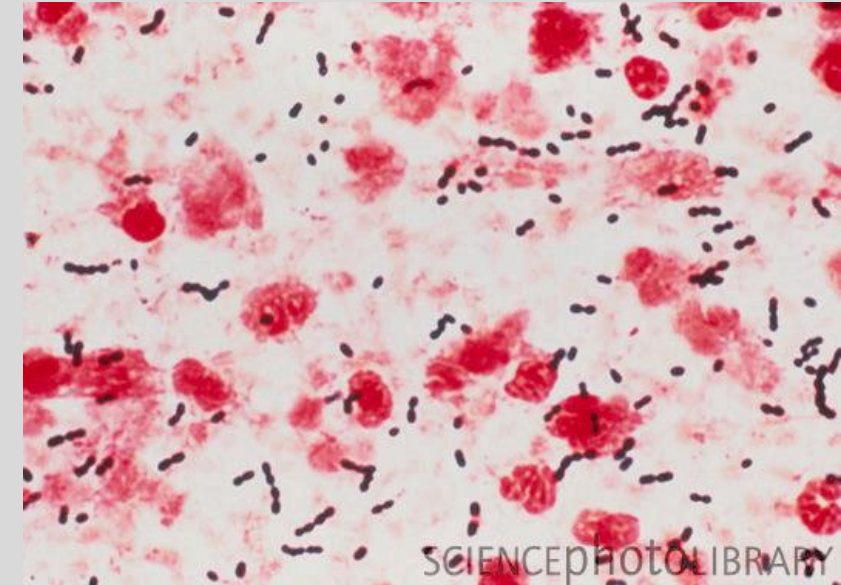
Acute Bacterial Meningitis

- Immediate treatment with antibiotics is necessary
 - **Remember CSF and blood cultures if accessible
- Start empirical, then choose based on possible etiology
 - Bactericidal beta-lactams (ceftriaxone or cefotaxime)
 - WITH vancomycin due to increasing resistance
 - Ampicillin for newborns (*Listeria*)
 - Dexamethasone (especially in *H. influenzae*)
- Chemoprophylaxis for immediate contacts
- Note: erythromycin, tetracycline, cefoxitin, 1st gen cephalosporins do not cross the BBB

Acute Bacterial Meningitis

- Etiological agents to know
 - *Streptococcus pneumoniae*
 - *Neisseria meningitidis*
 - *Haemophilus influenzae* type b
 - Group B *Streptococci* (*Streptococcus agalactiae*)
 - *Listeria monocytogenes*
 - *Escherichia coli*

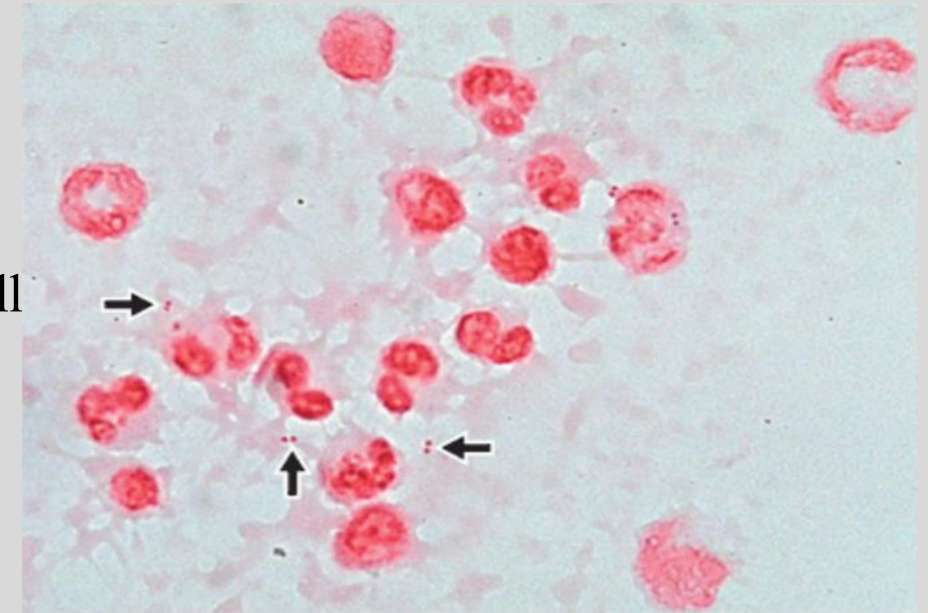
- *Streptococcus pneumoniae*
 - Gram +ve; lancet diplococci; α -hemolytic
 - Common inhabitant of nasopharynx and throat
 - Aerosol/droplet transmission
 - Purulent
 - 21% mortality
 - Mortality and severe neuro deficits are 4-20x more common than in other etiologies
 - >50% of isolates are beta lactam resistant
 - Capsule, IgA protease, pneumolytic
 - Capsular antigens can switch serotypes (genomic recombination/point mutations)
 - Pneumococcal Conjugate Vaccine (PCV)
 - 4 doses of PCV13 for infants followed by 1 dose of PCV23 8wks-1yr later; 1 dose of **PCV20** at 65yo and/or younger w/ risk factors
(<https://www.cdc.gov/vaccines/vpd/pneumo/hcp/who-when-to-vaccinate.html>)
- Asplenic patients/hypogammaglobinemia
- MOPS (**m**eningitis, otitis media, pneumonia, sinusitis)



Source: <http://vignette4.wikia.nocookie.net/mmg-233-2013-genetics-genomics/images/5/52/Sputum1.gif/revision/latest?cb=20131102221758>

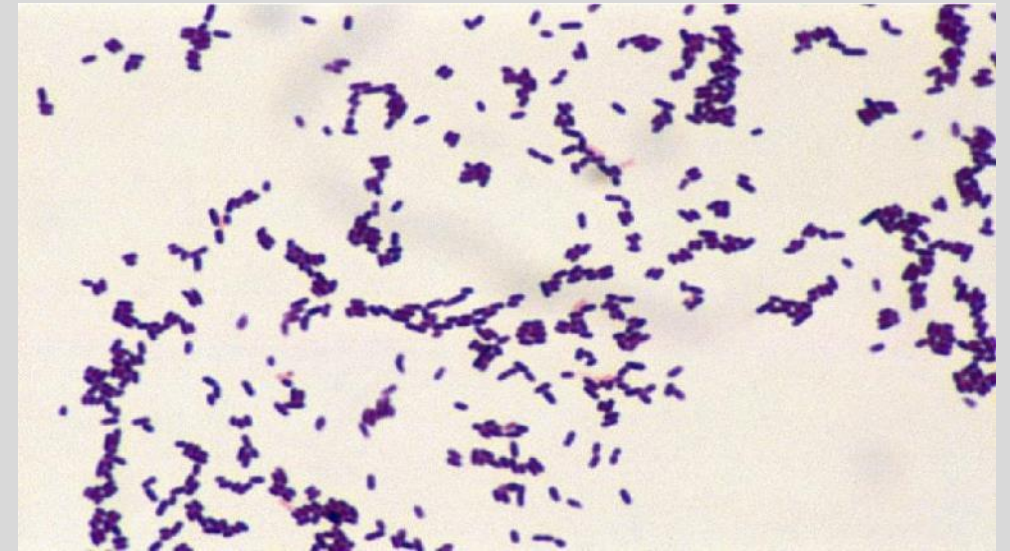
- *Neisseria meningitidis*

- Gram -ve; diplococci
- Aerosol/droplet transmission
- Colonizes 10% of healthy individuals
- Grows on blood agar
- Capsule, IgA protease, beta lactamase, release of blebs during cell growth
 - Antibodies to capsular Ag
- Complement plays a major role in defense
- Less neuro sequelae than other etiologies
- Tx with 3rd gen cephalosporins (used to be penicillin)
- Vaccine
 - 2 doses of MenACWY (covers strains A, C, W, Y) – one at 11 or 12yo, booster at 16yo
 - 1 dose of MenB (covers strain B) – at 16-23yo (<https://www.cdc.gov/vaccines/vpd/mening/index.html>)



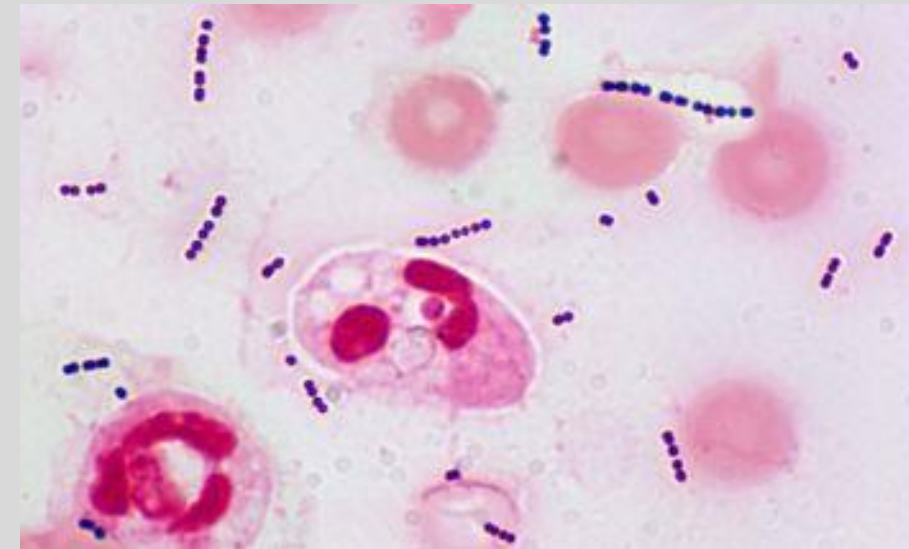
Source: Schaechter's **Mechanisms of Microbial Disease** 5th ed. Fig 61.4

- *Listeria monocytogenes*
 - Gram +vw; intracellular rod
 - 3rd most common cause of meningitis
 - Occurring in newborns (15% mortality), the elderly (20-50% mortality), the immunocompromised
 - Transplacental and fecal/oral transmission
 - Grows at 4°C (refrigeration temperature)
 - Tolerant to salts and nitrites (deli meat)
 - Listeriolysin, actin polymerization
 - Treat with ampicillin



Source: <http://www.msevens.com/cnsinfections/listeria-gs.jpg>

- Group B *Streptococci* (*Streptococcus agalactiae*)
 - Gram +vw; cocci in chains
 - Capsule, bacitracin resistant, β -hemolytic
 - Colonizes upper respiratory & genitourinary tracts
 - Acquired by neonates from mother during pregnancy or at time of birth
 - Higher risk of transmission with
 - Premature rupture of membranes
 - Prolonged labor
 - Preterm birth
 - Disseminated maternal Group B Strep
 - Mother doesn't have type specific antibodies or low complement
 - Penicillin G every 4hrs during labor



<https://microbe-canvas.com/Bacteria/gram-positive-cocci/streptococci/catalase-negative/beta-hemolysis/growth-on-mcconkey-agar-without-salt-1/streptococcus-agalactiae.html>

Review Question

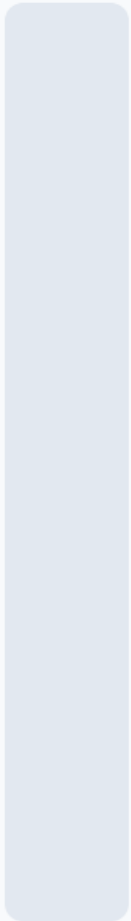
A 20-year-old male college student presents to the ED with nuchal rigidity, fever, and headaches for the past 3 days. He was accompanied by his roommate who claims the patient has had multiple periods of confusion since his symptoms began. His vaccination status is unknown, and he has no significant past medical history. A culture of his CSF is taken, and antibiotics are administered. Microscopic analysis of the CSF culture reveals gram negative diplococci.

What bacteria is the most likely etiology of this patient's meningitis?

Try identifying every clue in the question stem that led you to this answer.

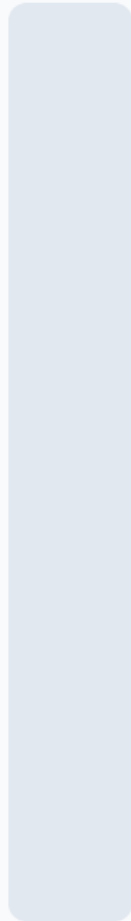
Which bacterium is the most frequent cause of meningitis in neonates?

0%



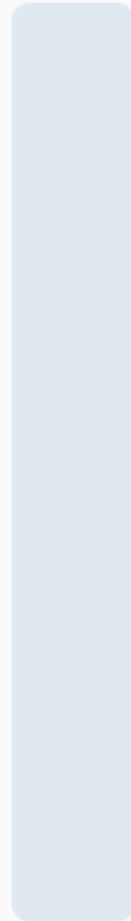
S. aureus

0%



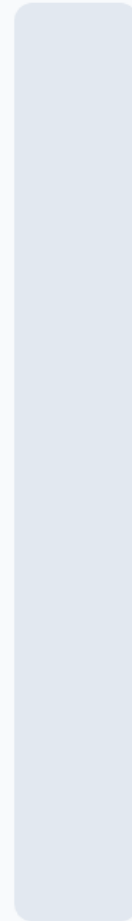
S. agalactiae

0%



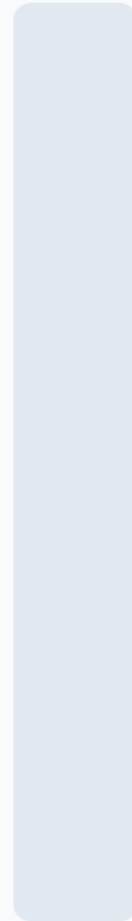
S. pneumoniae

0%



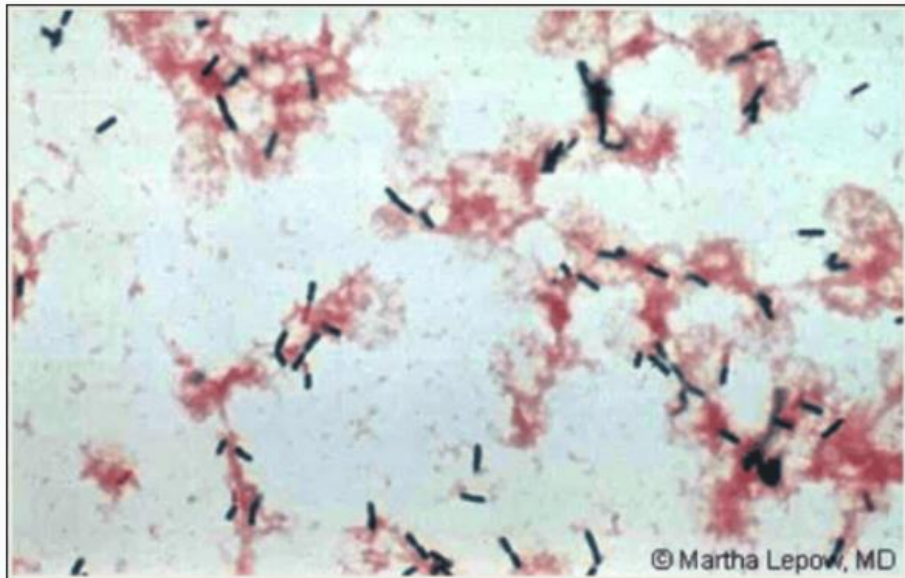
N. meningitidis

0%



L. monocytogenes

A 2-day-old male was admitted to the ER with classic signs of acute meningitis. CSF Gram stain micrograph is shown below. Which of the following is the likely etiological agent?



N. meningitidis

0%

S. agalactiae

0%

S. pneumoniae

0%

L. monocytogenes

0%

A 2-year-old female presents to the ER with signs of acute meningitis. Based on this information alone (and prior to gram stain CSF results), what is the best empirical antibiotic treatment?

Ampicillin + vancomycin

0%

Vancomycin + cefotaxime

0%

Erythromycin + cefoxitin

0%

Ampicillin + cefoxitin

0%

Dexamethasone + cefotaxime

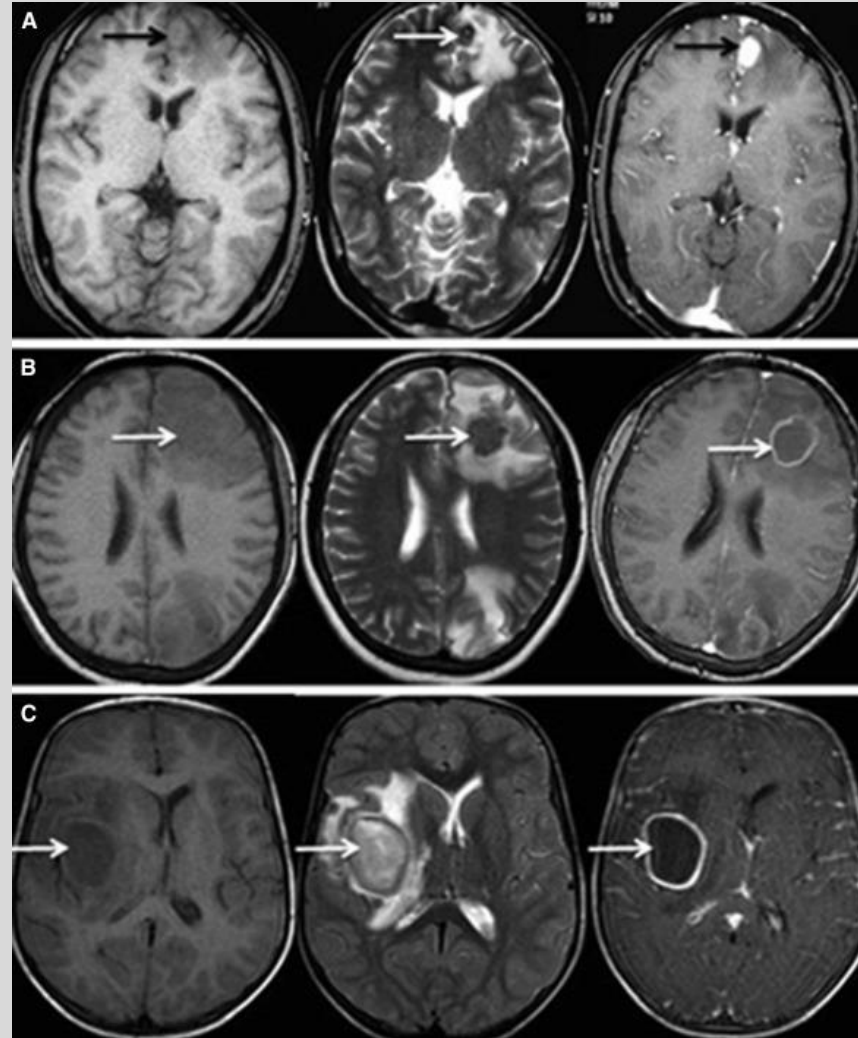
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Chronic Bacterial Meningitis

- Chronic – weeks to months of less severe symptoms
- Acute worsening can lead to seeking care
- Caused by fungi (*Cryptococcus neoformans*) and latent TB
 - Immunocompromised
 - Granuloma formation & inflammation
 - Cranial nerve dysfunction, increased pressure, coma
- Acid fast stain (*Rhodamine-Auramine*) and PCR of CSF do not rule out TB
- CSF culture for TB and presence of tuberculostearic acid in CSF

Chronic Bacterial Meningitis

- RIPE Tx
 - For TB
 - Rifampin, Isoniazid, Pyrazinamide, Ethambutol



Brain Abscesses

- Focal infection of parenchyma
- Symptoms include deficits of the affected tissue
 - Frontal lobe abscess – inability to focus or make sound decisions
 - Temporal lobe abscess – trouble speaking, auditory or visual deficits
- Entry
 - Direct inoculation from trauma or surgery
 - Hematogenous spread (septicemia or after dental work)
 - Spread from sinuses, middle ear, mastoid

Brain Abscesses

- Bacterial, fungal, parasitic etiologies
- Bacterial
 - Acute (polymicrobial)
 - *S. aureus*
 - Group A or D Strep
 - Mixed anaerobic and aerobic flora
 - Chronic
 - *Mycobacteria* (TB)

Brain Abscesses

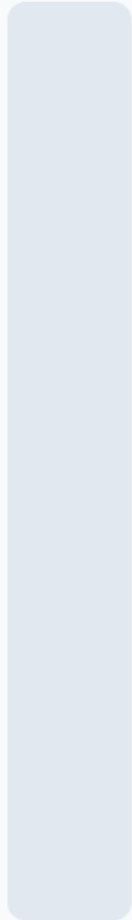
- Diagnosis
 - CT, MRI, aspiration
 - Rule out tumor or stroke
- Treatment
 - Antibiotics not always effective
 - Vancomycin (for methicillin-R)
 - Ceftriaxone, cefotaxime
 - Metronidazole
 - 10-20% mortality (fatal without tx)

Review Question

Without looking at the past slides, can you recall the ways to diagnose a case of chronic TB meningitis?

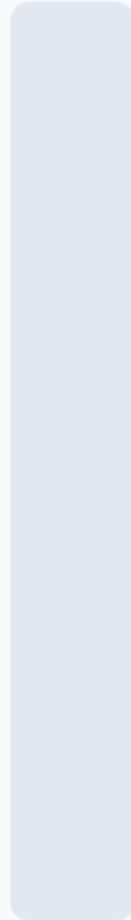
Which bacterium from the list below is the cause of chronic meningitis?

0%



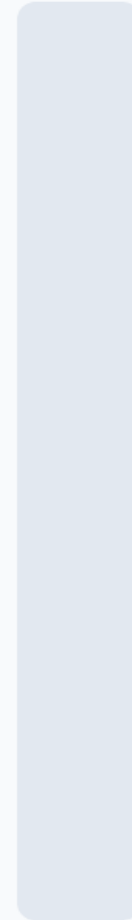
M. tuberculosis

0%



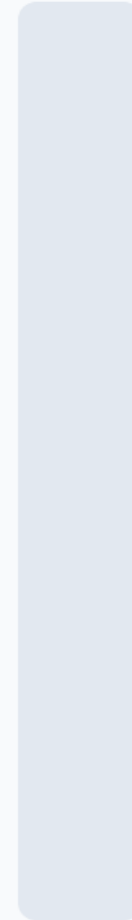
S. aureus

0%



S. pneumoniae

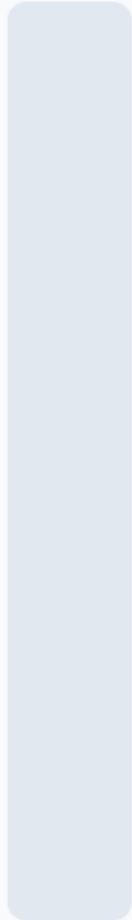
0%



S. agalactiae

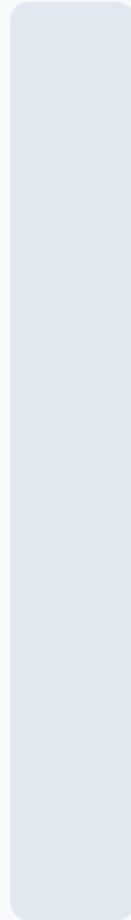
Which bacterium from the list below is the cause of acute brain abscesses?

0%



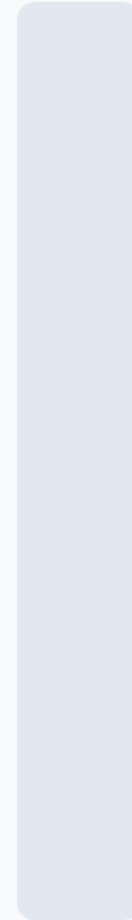
M. tuberculosis

0%



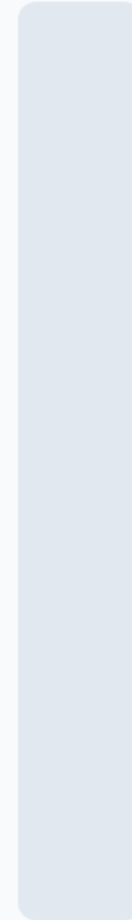
S. aureus

0%



S. pneumoniae

0%



S. agalactiae

Summary Slide

- Limited access to CNS by microbes
- CNS bacterial infections: meningitis and abscess
- Often severe to life threatening
- Immediate action (relief pressure, stabilize, draw CSF, empirical antibiotic)
- CSF microbiology (collect prior to antibiotic administration) and chemistry evaluation is crucial
- Vaccination!

References

N. C. Engleberg, V. J. DiRita and M. J. Imperiale. 2021. **Schaechter's Mechanisms of Microbial Disease** 6th ed.; Lippincott Williams & Wilkins, a Wolters Kluwer business. Ch. 60

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Thank you!!!



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